

STQD6114

TEXT DATA ANALYSIS III:

SENTIMENT ANALYSIS



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Introduction

- ❖ Sentiment: a view, an opinion
- ❖ Sentiment analysis: a process of computationally identifying and categorizing sentiments typically expressed in a text
- ❖ Determining emotional tone behind a series of word



Why?

- ❖ Social media monitoring – gain overview on public opinions for a certain topic
- ❖ Able to quickly understand consumer needs and react to it
- ❖ Example: Expedia Canada commercial case



Loves the German bakeries in Sydney. Together with my imported honey it feels like home	Positive
@VivaLaLauren Mine is broken too! I miss my sidekick	Negative
Finished fixing my twitter...I had to unfollow and follow everyone again	Negative
@DinahLady I too, liked the movie! I want to buy the DVD when it comes out	Positive
@frugaldougal So sad to hear about @OscarTheCat	Negative
@Mofette brilliant! May the fourth be with you #starwarsday #starwars	Positive
Good morning thespians a bright and sunny day in UK, Spring at last	Positive
@DowneyisDOWNEY Me neither! My laptop's new, has dvd burning/ripping software but I just can't copy the files somehow!	Negative

From comment, identify the sentiment based on the language



Jadual 4.6 Perbandingan Kutipan Filem dengan Markah Akhir Kajian Filem Antarabangsa Tahun 2015

Filem (Antarabangsa)	Kutipan (USD)*	Bilangan Positif	Bilangan Negatif	Markah Akhir
<i>Avengers: Age of Ultron</i>	1,405,413,868	542	94	85.22
<i>Furious 7</i>	1,516,045,911	529	101	83.97
<i>Jurassic World</i>	1,670,400,637	764	248	75.49
<i>Minions</i>	1,159,398,397	448	406	52.46
<i>Star Wars: The Force Awakens</i>	2,066,960,090	303	132	69.66



Hotel	Skor sentimen keseluruhan Agoda	Skor penarafan Agoda	Skor sentimen keseluruhan Booking.com	Skor penarafan Booking.com
One World	6.85	8.5	6.59	8.5
Pullman Putrajaya	6.6	8	6.66	8.2
Vibrant Studio	6.87	8.6	8.23	8.8
Golden Triangle	6.37	7.1	6.47	7.1
The Gardens	6.83	8.3	6.79	8.4
Ascott Kuala Lumpur	6.57	8.5	6.64	8.7
Summer Suite	7.27	8.4	7.1	8.7
Sarang Vacation	7.1	8.6	6.8	9.1

Penarafan

Vibrant Studio	6.87	7
Golden Triangle	6.37	7
The Gardens	6.83	10
Ascott Kuala Lumpur	6.57	9
Summer Suite	7.27	8
Sarang Vacation	7.1	6

ntimen keseluruhan dengan skor bintang

Skor sentimen keseluruhan Booking.com	Skor Bintang Booking.com
6.59	10
6.66	10
8.23	0
6.47	0
6.79	10
6.64	10
7.1	0
6.8	0

- ❖ Teaching machine to identify context and sentiment of human language is very difficult
- ❖ Human language itself is already complex , and add on the lack of intuitively in a machine: how can we do it?
- ❖ Example: Wow, Astro doesn't broadcast when its rain! Verrrryyyyy goooooddd!!
- ❖ A human know that the above sentence need to be read in a sarcastic way; hence it is a negative tone, however a machine sees the word "good" & might categorize as a positive tone statement
- ❖ Hence, the algorithm is evolving (as we talk!) to include comprehensively phrases/statements to increase the ability of a machine in conducting the sentiment analysis.



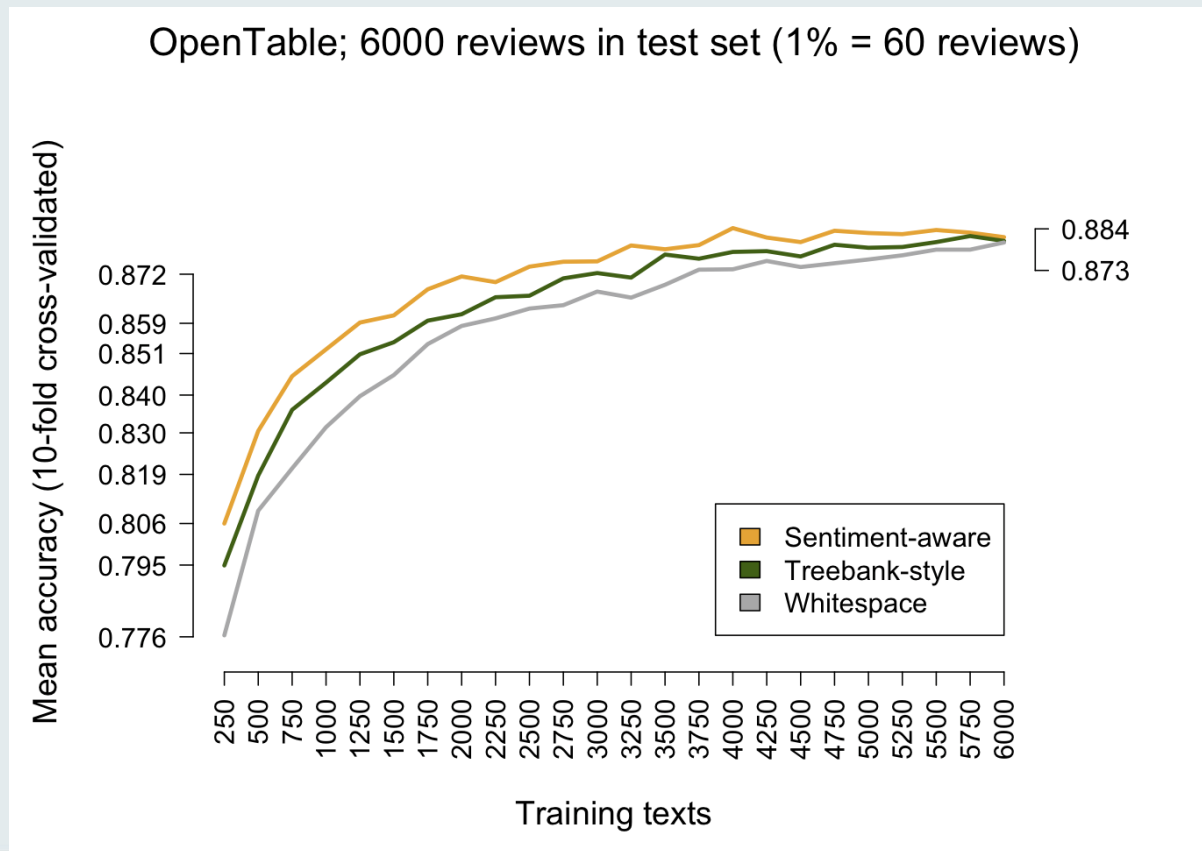
- ❖ Hence, the sentiment analysis results needs to be taken 'with a pinch of salt' (in precaution; with warning)
- ❖ It is not 100% accurate (yet!) but it do provides the overview / general idea especially on public sentiment





Data Preprocessing (Cleaning & Scrubbing)

- ❖ Text input
- ❖ Tokenization: splitting a string into its desired constitutes parts.



- ❖ Stop word filtering
- ❖ Negation handling: not good, not not good; not pretty, not not pretty, pretty ugly?
- ❖ Stemming



Data Sentiment Analysis

- ❖ Classification: classifying positive, negative words
- ❖ Sentiment class: determine the polarity of the topic or the data



How to determine the sentiment score

❖ Capitalization

- Words that are capitalized often signify a stronger expression - *Angry, scream*

❖ Emotion

- Usage of emotions alone
- Usage of emotions along with the words

❖ The length of phrase

- Longer phrases - *high score of -ve or +ve*
- *high detail*
- Repetition on synonymous words

❖ Examples:

- I am beautiful *normal*
- I am very beautiful
- I am BEAUTIFUL *caps*
- I am superrrr beautiful *strong sentiment*
- Beautifullllllllll



Sentiment Analysis in R (Simple codes)

Data prepping:

```
library(tm)
library(SnowballC)
library(wordcloud)
text=readLines(file.choose())
docs=Corpus(VectorSource(text))
inspect(docs)
toSpace=content_transformer(function(x,pattern)gsub(pattern," ",x))
docs=tm_map(docs, toSpace, "/")
docs=tm_map(docs, toSpace, "@")
docs=tm_map(docs, toSpace, "\\|")
docs=tm_map(docs,content_transformer(tolower))
docs=tm_map(docs,removeNumbers)
docs=tm_map(docs,removeWords, stopwords("english"))
docs=tm_map(docs,removeWords, c("dan","dengan","atau","sebagai","yang","itu", "ini","asm","dari","daripada"))
docs=tm_map(docs,removePunctuation)
docs=tm_map(docs,stripWhitespace)
docs=tm_map(docs,stemDocument)
dtm=TermDocumentMatrix(docs)
m=as.matrix(dtm)
v=sort(rowSums(m),decreasing=TRUE)
d=data.frame(word=names(v),freq=v)
m=d$word
```



Sentiment Analysis in R

Sentiment analysis:

```
mysentiment<-function(m)
{
  mydictpos=c("baik", "cantik", "bijak", "kuat")
  mydictneg=c("jahat", "buruk", "bodoh", "lemah")
  pos_score=sum(!is.na(match(m,mydictpos)))
  neg_score=(-1)*sum(!is.na(match(m,mydictneg)))
  sentiment_score=pos_score+neg_score
  sentiment_score
}
```



Sentiment Analysis by lexicon

Lexicon means dictionary

Example: affin, bing ,syuzhet ,nrc

calculate sentiment
score for every
lexicon

affin

```
## # A tibble: 2,477 x 2
##   word      value
##   <chr>    <dbl>
## 1 abandon      -2
## 2 abandoned    -2
## 3 abandons     -2
## 4 abducted     -2
## 5 abduction    -2
## 6 abductions   -2
## 7 abhor        -3
## 8 abhorred     -3
## 9 abhorrent    -3
## 10 abhors      -3
## # ... with 2,467 more rows
```

bing

```
## # A tibble: 6,786 x 2
##   word      sentiment
##   <chr>    <chr>
## 1 2-faces  negative
## 2 abnormal negative
## 3 abolish negative
## 4 abominable negative
## 5 abominably negative
## 6 abominate negative
## 7 abomination negative
## 8 abort    negative
## 9 aborted  negative
## 10 aborts  negative
## # ... with 6,776 more rows
```

every word has score



Sentiment Analysis by lexicon in R

We will refer to:

<https://www.tidyttextmining.com/sentiment.html>

Another example:

https://rstudio-pubs-static.s3.amazonaws.com/302066_fe1dd2a635fa41198b18c87a64f5620c.html



Appendix

```
library(tm) # for text mining
library(SnowballC) # for text stemming
library(wordcloud) # word-cloud generator
library(RColorBrewer) # color palettes
library(syuzhet) # for sentiment analysis
library(ggplot2) # for plotting graphs
```

```
# Read the text file from local machine , choose file interactively
text <- readLines(file.choose())
```

```
# Load the data as a corpus
docs <- Corpus(VectorSource(text))
```

```
#Replacing "/", "@" and "|" with space
toSpace <- content_transformer(function (x , pattern ) gsub(pattern, " ", x))
docs <- tm_map(docs, toSpace, "/")
docs <- tm_map(docs, toSpace, "@")
docs <- tm_map(docs, toSpace, "www|")
```

```
docs <- tm_map(docs, content_transformer(tolower))
docs <- tm_map(docs, removeNumbers)
docs <- tm_map(docs, removeWords, stopwords("english"))
docs <- tm_map(docs, removeWords, c("s", "company", "team"))
docs <- tm_map(docs, removePunctuation)
docs <- tm_map(docs, stripWhitespace)
docs <- tm_map(docs, stemDocument)
```

For wordcloud



Appendix

```
# Build a term-document matrix
dtm <- TermDocumentMatrix(docs)
dtm_m <- as.matrix(dtm)
dtm_v <- sort(rowSums(dtm_m),decreasing=TRUE) # Sort by descending value of frequency
dtm_d <- data.frame(word = names(dtm_v),freq=dtm_v)
head(dtm_d, 5) # Display the top 5 most frequent words

# Plot the most frequent words
barplot(dtm_d[1:5,]$freq, las = 2, names.arg = dtm_d[1:5,]$word,
        col = "lightgreen", main = "Top 5 most frequent words",
        ylab = "Word frequencies")

#generate word cloud
set.seed(1234)
wordcloud(words = dtm_d$word, freq = dtm_d$freq, min.freq = 5,
          max.words=100, random.order=FALSE, rot.per=0.40,
          colors=brewer.pal(8, "Dark2"))

# Word Association
findAssocs(dtm, terms = c("good","work","health"), corlimit = 0.25) # Find associations
findAssocs(dtm, terms = findFreqTerms(dtm, lowfreq = 50), corlimit = 0.25) # Find associations for words that occur at least 50 times
```



Appendix

Sentiment scores

regular sentiment score using get_sentiment() function and method of your choice

please note that different methods have different scales

```
syuzhet_vector <- get_sentiment(text, method="syuzhet")
```

```
head(syuzhet_vector)
```

```
head(syuzhet_vector,10) # see the first 10 elements of the vector
```

```
summary(syuzhet_vector)
```

bing

```
bing_vector <- get_sentiment(text, method="bing")
```

```
head(bing_vector)
```

```
summary(bing_vector)
```

#afinn

```
afinn_vector <- get_sentiment(text, method="afinn")
```

```
head(afinn_vector)
```

```
summary(afinn_vector)
```

#nrc

```
nrc_vector <- get_sentiment(text, method="nrc")
```

```
head(nrc_vector)
```

```
summary(nrc_vector)
```

#compare the first row of each vector using sign function
rbind(
 sign(head(syuzhet_vector)),
 sign(head(bing_vector)),
 sign(head(afinn_vector))
)



Appendix

```
## Emotion classification
```

```
# run nrc sentiment analysis to return data frame with each row classified as one of the following
```

```
# emotions, rather than a score :
```

```
# anger, anticipation, disgust, fear, joy, sadness, surprise, trust
```

```
# and if the sentiment is positive or negative
```

```
d<-get_nrc_sentiment(text)
```

```
head (d,10) # head(d,10) - just to see top 10 lines
```

```
#Visualization
```

```
td<-data.frame(t(d)) #transpose
```

```
td_new <- data.frame(rowSums(td)) #The function rowSums computes column sums across rows for each level of a grouping variable.
```

```
names(td_new)[1] <- "count" #Transformation and cleaning
```

```
td_new <- cbind("sentiment" = rownames(td_new), td_new)
```

```
rownames(td_new) <- NULL
```

```
td_new2<-td_new[1:8,]
```

```
#Plot 1 - count of words associated with each sentiment
```

```
quickplot(sentiment, data=td_new2, weight=count, geom="bar",fill=sentiment,ylab="count")+ggtitle("Survey sentiments")
```

```
#Plot 2 - count of words associated with each sentiment, expressed as a percentage
```

```
barplot(
```

```
  sort(colSums(prop.table(d[, 1:8]))),
```

```
  horiz = TRUE,
```

```
  cex.names = 0.7,
```

```
  las = 1,
```

```
  main = "Emotions in Text", xlab="Percentage"
```

```
)
```

